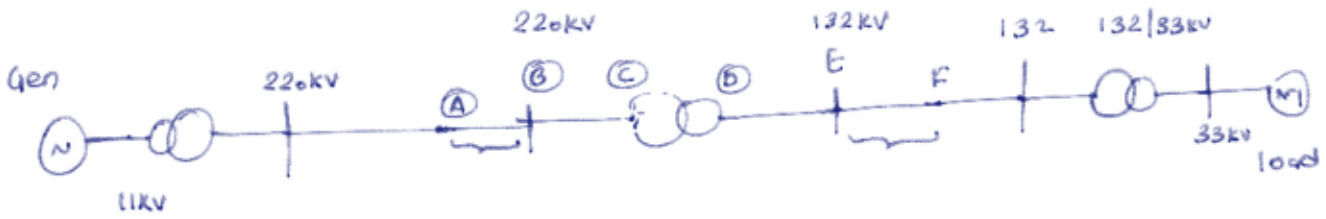
Protection Basics

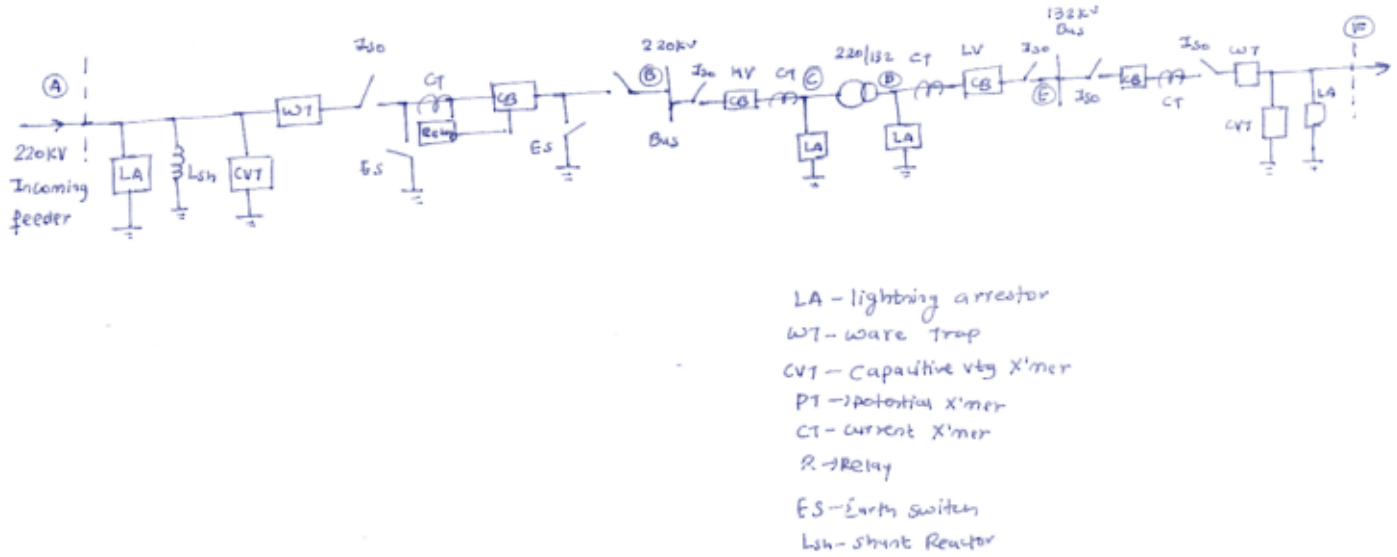
Switchgear and Protection :

P.S N/W consisting of Generator, Transformers, T/L, Bus-bars, motors, power cables etc. All these equipment having certain power, current, vtg withstand capability. If any abnormal condition like S.C and O.C fault occurs in the system, and lightning and outage of equipments induce huge current and vtg into the system which damages the electrical appliances. To protect these equipments, protecting devices like isolator, C.B, Earth switch, CT and PT are arranged in specific sequences in each phase of the S/S equipments as per the Indian electricity grid code.



Generator, 11 kV step-up transformer, 220 kV bus, points A, B, C, D (220/132 kV transformer), E, F on 132 kV, 132/33 kV transformer, 33 kV motor load

(1) During healthy normal supply condition, Earth switch must be in open condition and Iso and CB are in closed condition.



Single-line diagram of a substation, from point A (220 kV incoming feeder) to point F: LA lightning arrester, Lsh shunt reactor, CVT, WT wave trap, Iso isolator, ES earth switch, CT, relay, CB, 220 kV bus (B), HV CT, 220/132 kV transformer (C), LV CT, CB, Iso, 132 kV bus (E), WT, CVT, LA

LA - lightning arrester ; WT - wave trap ; CVT - Capacitive vtg X'mer ; PT - Potential X'mer ; CT - Current X'mer ; R - Relay ; ES - Earth switch ; Lsh - shunt Reactor

(1) LA - Lightning arrester

* First and last protecting device of the S/S is LA, which is made of metal oxides like Zinc oxide (ZnO), which offers low impedance for the surge vtg and high impedance for the normal supply vtg.
* 2 more LA are placed at both ends of the power X'mer in each phase for protection of T/F from the lightning surges occurring within the S/S.

(2) Lsh (Shunt Reactor)

Shunt reactors are placed to protect S/S equipments from the over vtg that occurs due to Ferranti effect.

(3) CVT

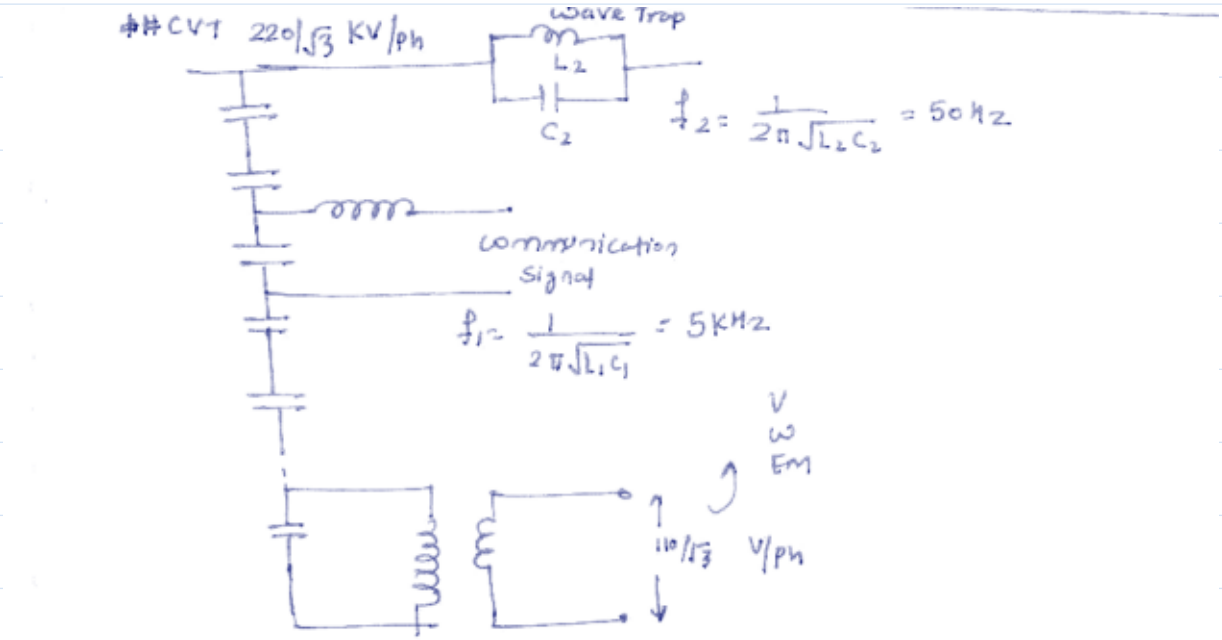
It works on the principle of capacitance, used to measure vtg of each phase of the feeder, and this vtg signal is applied to voltmeter, potential coil of wattmeter and energy meter and protective relays.

It has internally an LC tuned circuit, tuned for communication frequency used for Power Line Carrier Communication (PLCC).

(4) Wave Trap / Line Trap

It consists of an LC tuned circuit, tuned for power frequency, and it offers low impedance for the power frequency signal and high impedance for the communication signal.

CVT $220/\sqrt{3}$ kV/ph



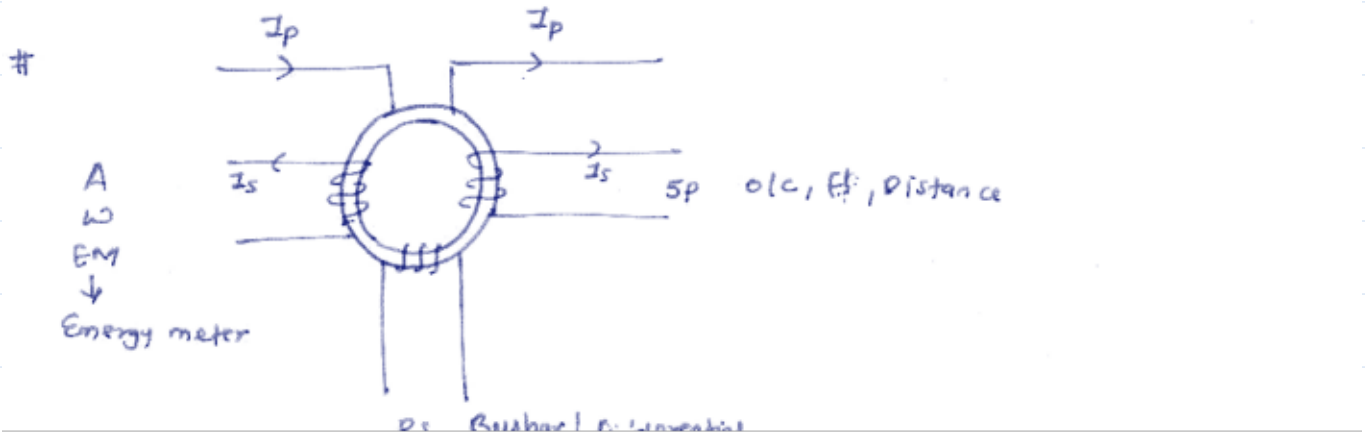
Wave trap L2 // C2 with f2 = 1/(2 pi sqrt(L2 C2)) = 50 Hz; capacitor stack of CVT with communication signal tapped off, f1 = 1/(2 pi sqrt(L1 C1)) = 5 kHz; electromagnetic unit giving 110/sqrt3 V/ph

CT : Current X'mer

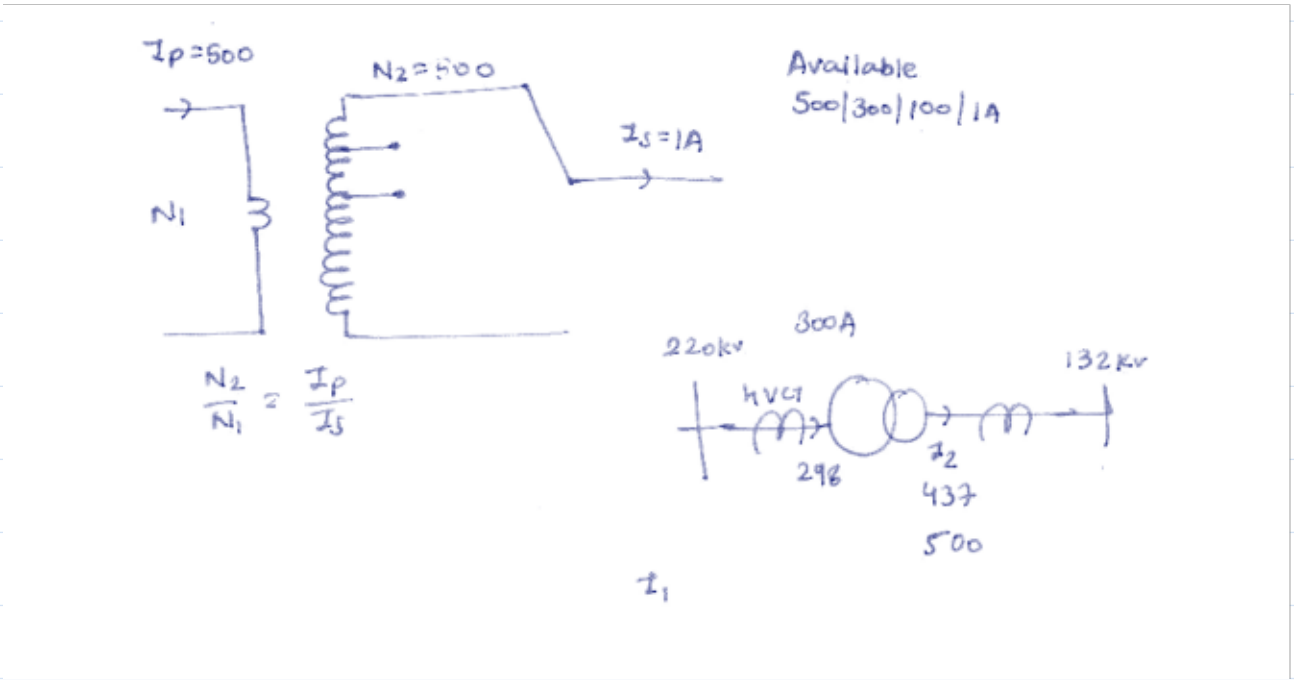
* CT has single turn primary and multiple secondary turns and also secondary winding.
* It is used to measure the large current in the feeder, X'mer, T/L, Generator etc.

* It provides the current signal to the ammeter, wattmeter and protective relays like O.C, Distance, Differential etc.

* Standard secondary current of CT is 1 A or 5 A.



Ring-core CT with primary Ip passing through; secondary cores: A, W, EM (energy meter); 5P cores for OC, EF, Distance; PS core for busbar / differential



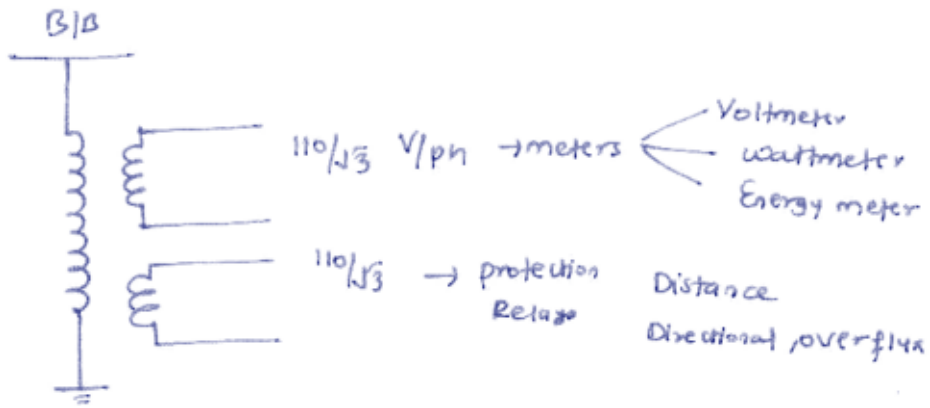
CT with Ip = 500 A, N1 primary, N2 = 500 secondary turns, Is = 1 A; available ratios 500/300/100/1 A; example 220 kV / 132 kV transformer with HV and LV CTs, currents 300 A, 298, 437, 500

$$\frac{N_2}{N_1} = \frac{I_p}{I_s}$$

PT (Potential X'mer) :

* It is used to measure vtg in each phase of the busbar.

* Standard secondary vtg of the PT is $110/\sqrt{3}$ V/ph. It provides vtg signal to voltmeter, wattmeter, energy meters and protective relays like Distance, Directional, overflux, etc.



Bus (B/B) PT with two secondaries each 110/sqrt3 V/ph: one to meters (voltmeter, wattmeter, energy meter), one to protection relays (distance, directional, overflux)

Circuit Breaker :

* It consists of an arc quenching medium so that it can be opened or closed either on no load, full load or fault conditions.

Isolator :

It must be opened or closed under no load condition only because it doesn't contain any arc quenching medium.

Earth Switch :

* It is used to discharge residual energy present in the electrical appliances so that it can be used during maintenance period.

The following sequence is used before attending maintenance :

(1) Open CB

(2) Open Isolator

(3) Close Earth Switch